

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- Midterm revisions due on Friday.
- This week: Last thoughts on Chapter 4; Chapter 5
- HW₅ (written and WeBWorK) due on tonight at 11:59pm.
- Requizzing session today, Quizzes 1-8, 3:30-4:30pm in Eckhart 207A
- Problem swap 2 problems due on Wednesday.
- Quiz 9 on Friday (eigenvalues and eigenvectors.)

For eigenvectors and eigenvalues, diagonal matrices are nice!

For a given operator, is there a basis where the corresponding matrix is diagonal?

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1. Find the eigenvalues and eigenvectors of A .
2. If geometric multiplicity = algebraic multiplicity, proceed! Else, can't be diagonalized.
3. Use “change of basis” matrices to write $A = QDQ^{-1}$, where D has the eigenvalues of A on the diagonal and Q has the corresponding eigenvectors as its columns.

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$$\lambda_1 = 1, v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$$
$$\lambda_2 = 2, v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

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$$\lambda_1 = 1, v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} = Q \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} Q^{-1}$$
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Diagonalization is useful
for taking powers of matrices.

$$\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}^8 = \left(\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \right)^8$$

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$$= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 256 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1}$$

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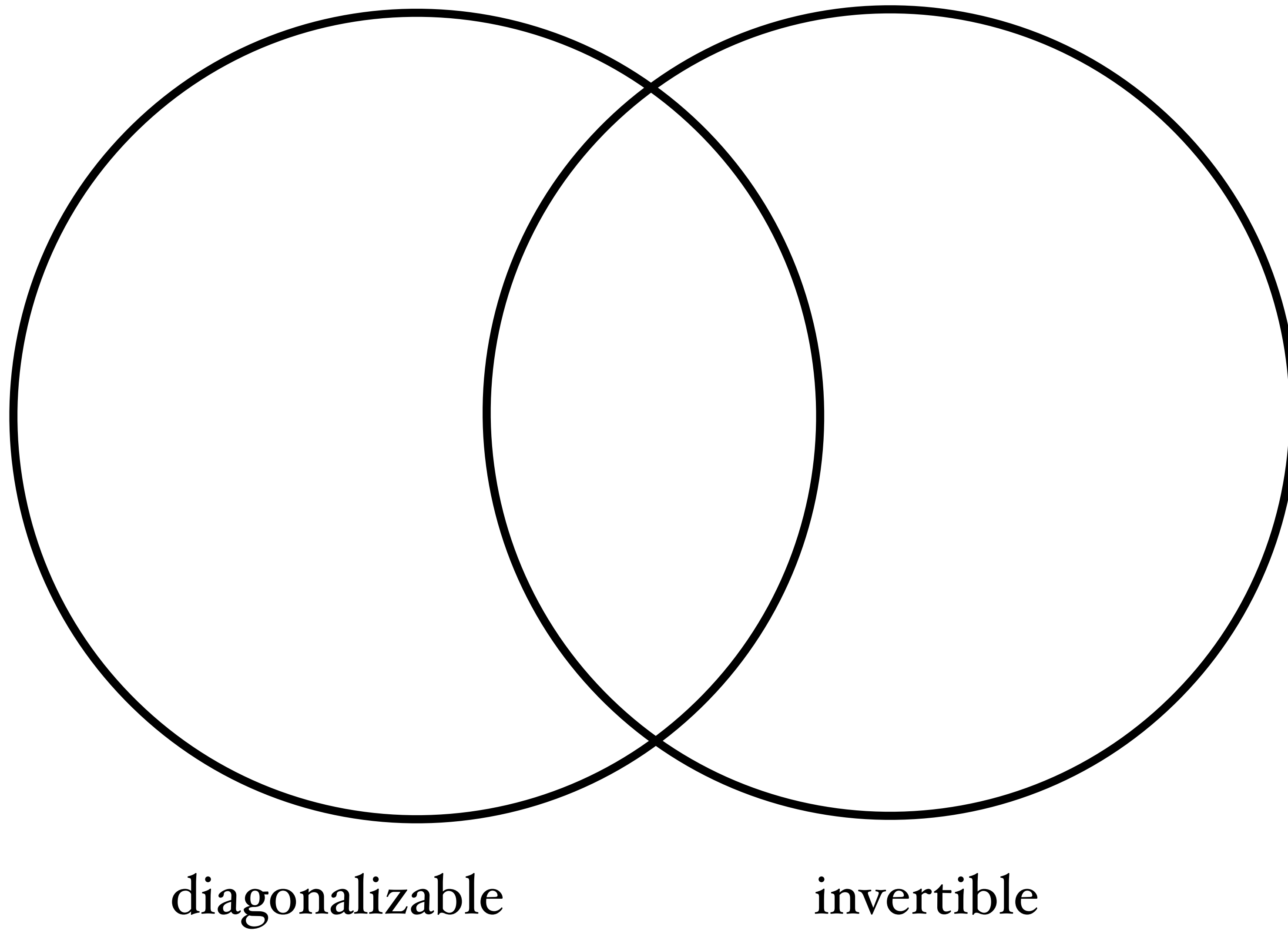
$$= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^8 \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1}$$

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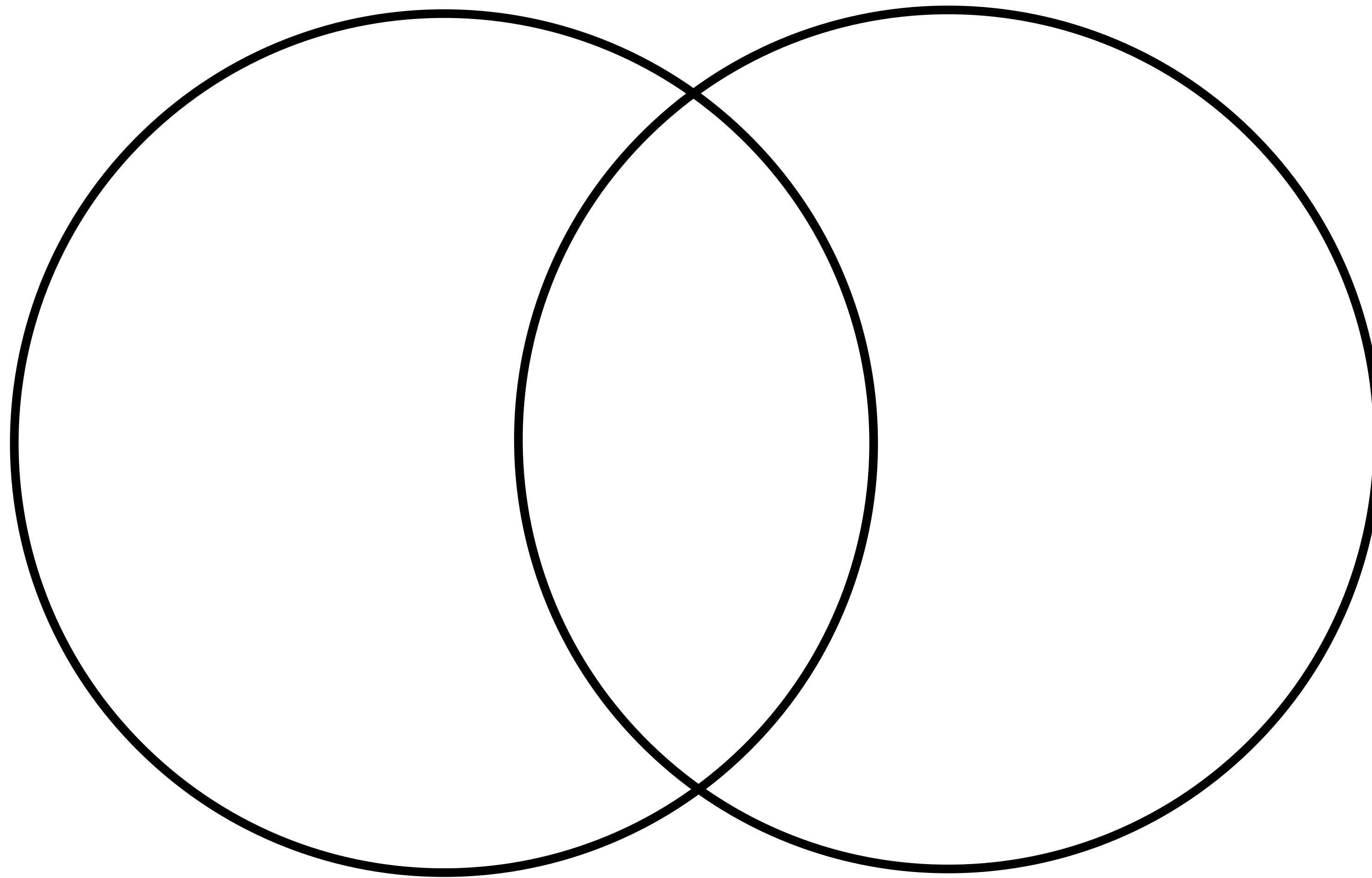
$$= \begin{pmatrix} -254 & 255 \\ -510 & 511 \end{pmatrix}$$

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How are diagonalizability and invertibility related?



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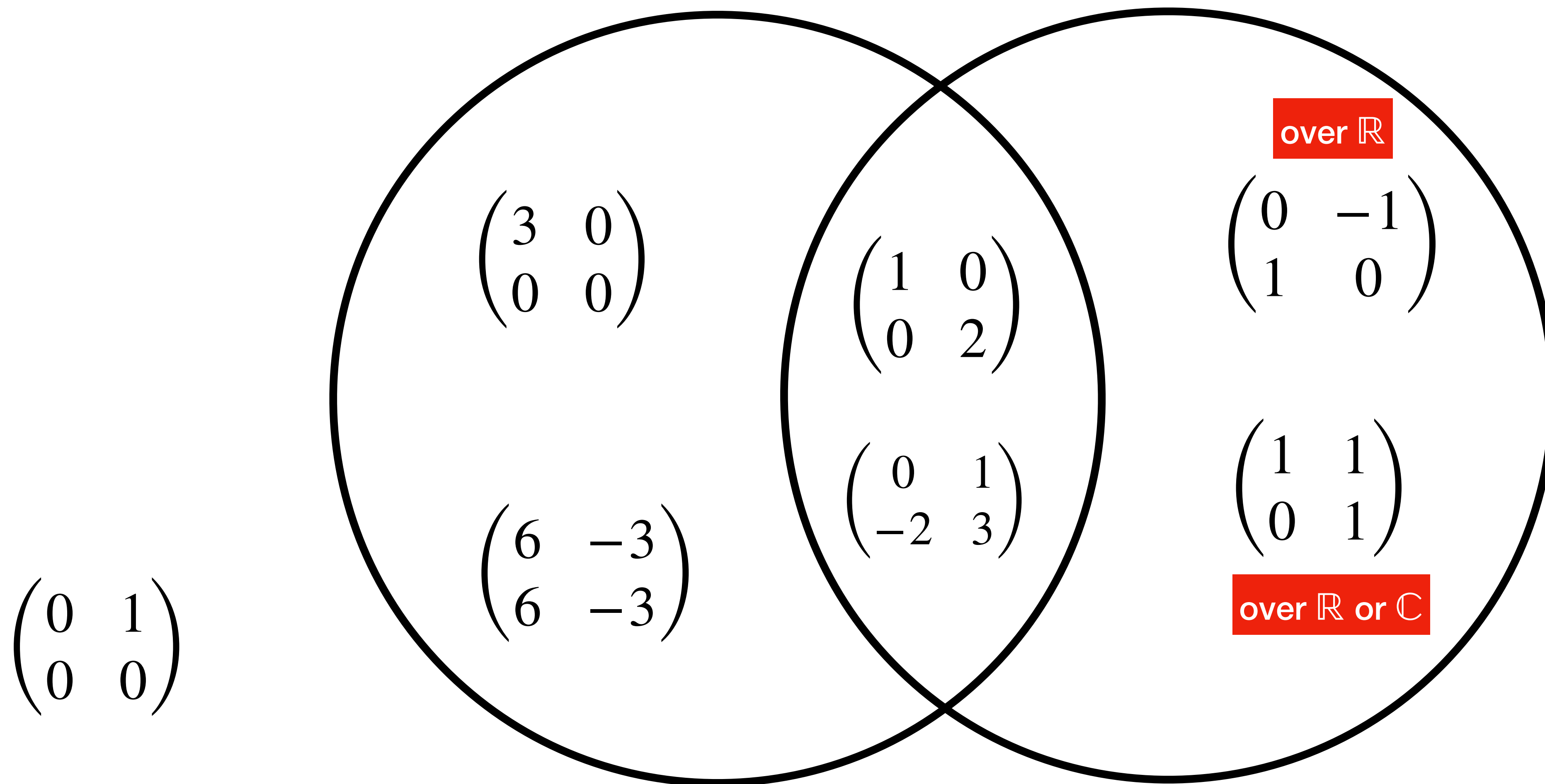
diagonalizable

- Includes all diagonal matrices.
- To be diagonalizable, we must be able to form a basis for V with eigenvectors of A .

invertible

- We have many criteria.
- In particular, A must have no eigenvalues equal to 0.

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Approaches to finding eigenvalues:

- Compute characteristic polynomial, find roots
- Finding roots of general polynomials is hard and often requires iterative methods.

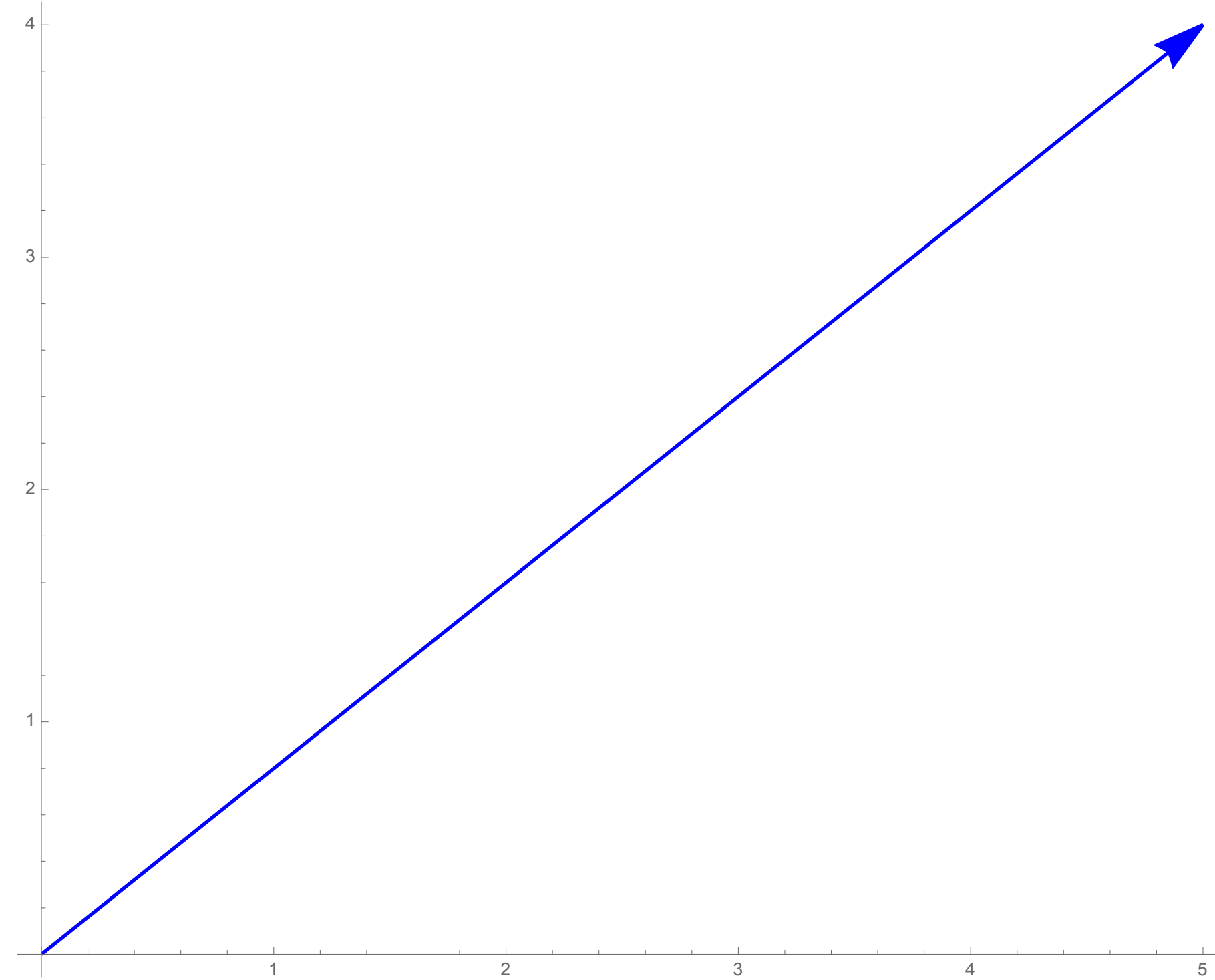
Newton's method

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- Iterative methods:
 - Power algorithm
 - QR algorithm

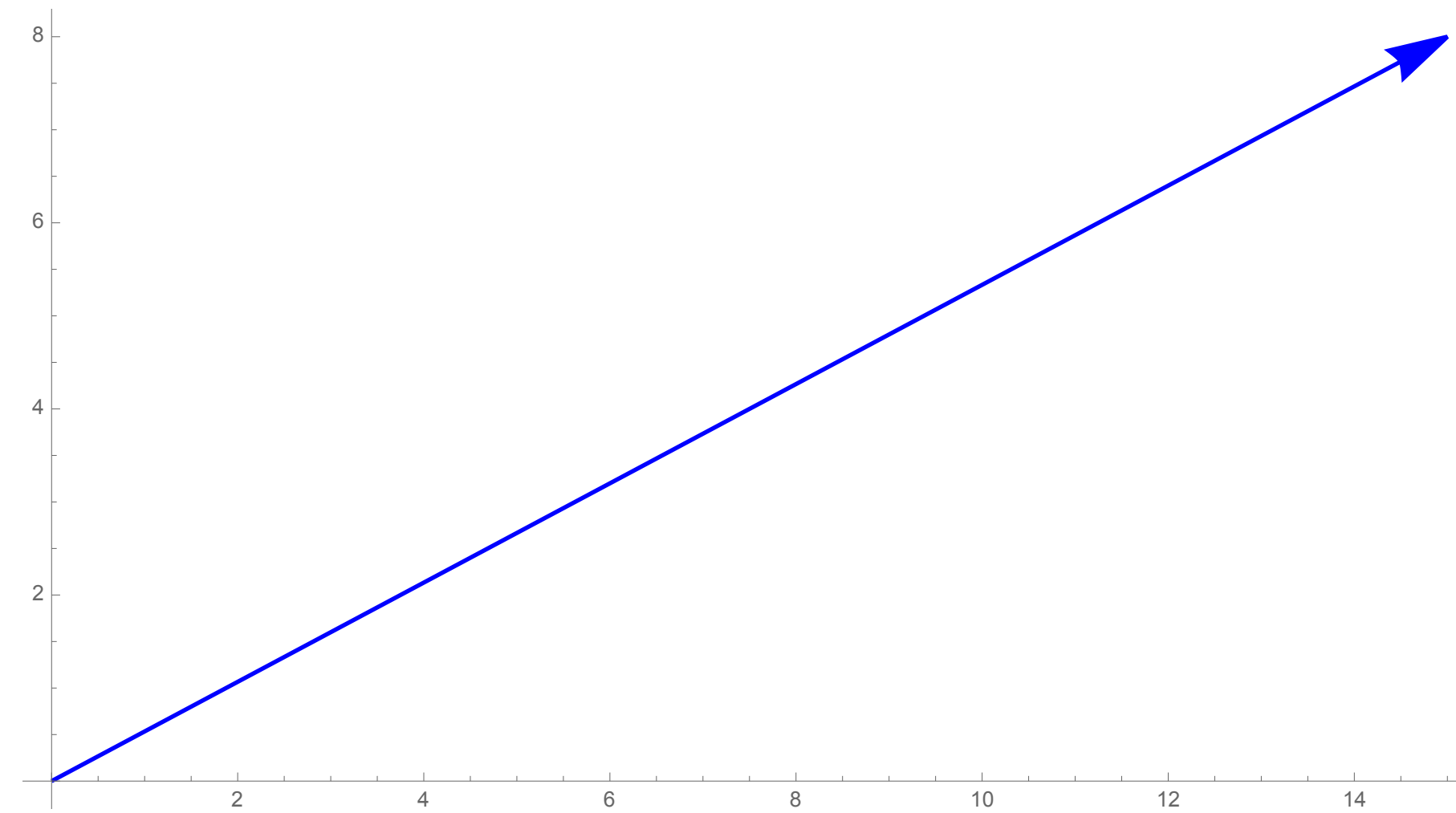
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Power method:



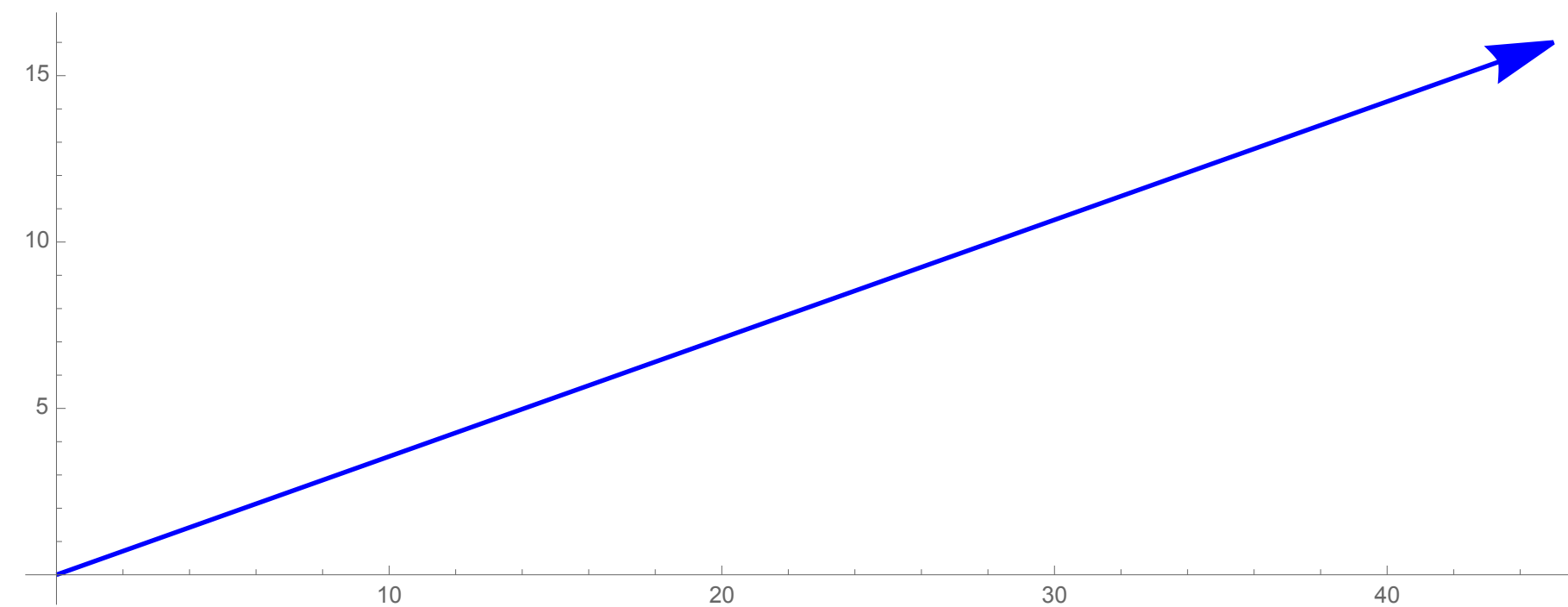
$$\begin{pmatrix} 5 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 4 \end{pmatrix}, \quad \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}^2 \begin{pmatrix} 5 \\ 4 \end{pmatrix}, \quad \dots$$

Power method:



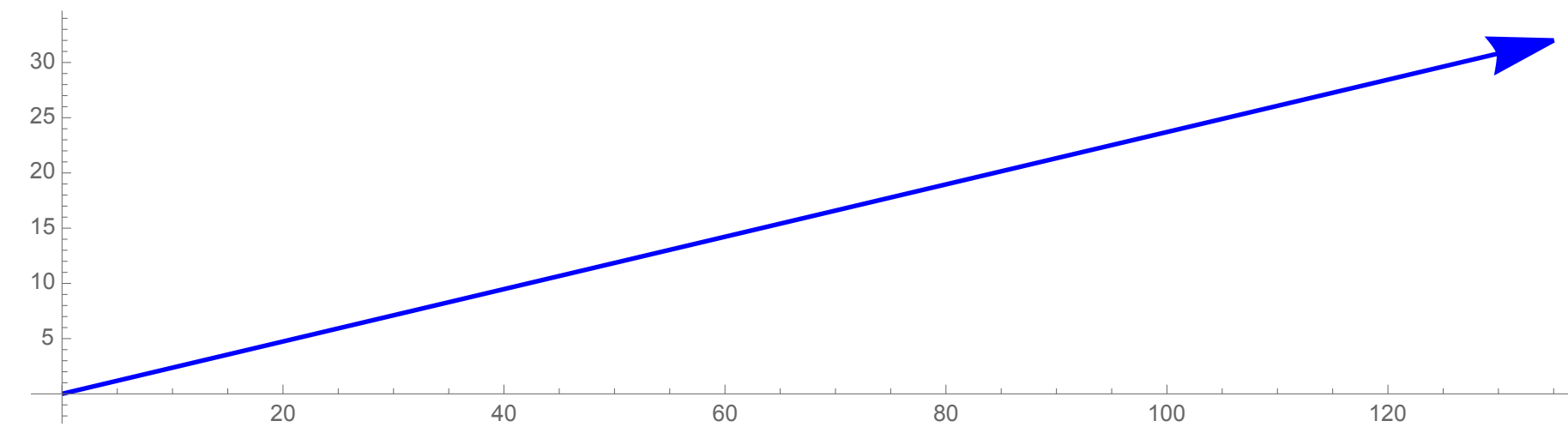
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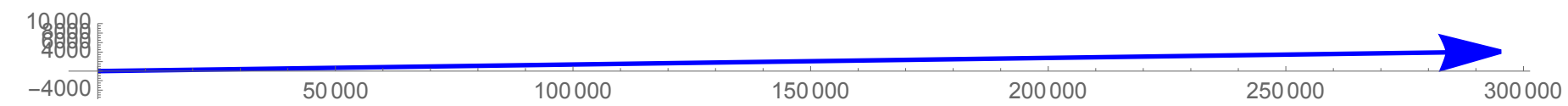
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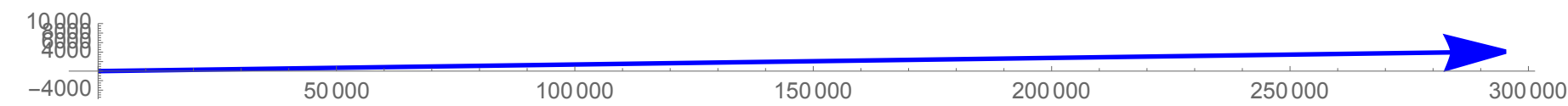
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Power method:

Iterating an operator on an arbitrary starting vector
“finds” the eigenspace with the largest eigenvalue.



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month	babies	adults
0	I	O
I	O	I
2		
3		
4		
...		



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...		



In the long term:
- At what rate is the population growing?
-How does the ratio of babies to adults evolve?

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transition matrix



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$$\det \begin{pmatrix} -\lambda & 1 \\ 1 & 1 - \lambda \end{pmatrix}$$

$$\lambda^2 - \lambda - 1 = 0$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$



$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad \lambda = \frac{1 \pm \sqrt{5}}{2}$$

$$\frac{1 + \sqrt{5}}{2} \text{ eigenspace} = \text{span} \left(\begin{pmatrix} 1 \\ \frac{1 + \sqrt{5}}{2} \end{pmatrix} \right)$$

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$$\Rightarrow F_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n \quad \text{closed form!}$$

Ch 5

inner products and orthogonality



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with just its bare vector space structure.

Ch 5

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But by layering on additional structure—an inner product—
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Ch 5

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Reminder:
going forward we are
only thinking about vector
spaces over $\mathbb{R}$ and $\mathbb{C}$.



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A multilinear form on a vector space F^n is a function on k -tuples of vectors (for some fixed k) to the field F that is linear in each input/argument.

Examples:

The determinant

The dot product (a symmetric bilinear form)

The dot product of u and v is just $u^T v$ with usual matrix multiplication. It is useful to consider the result as a field element, rather than as a matrix. (standard inner product)

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

$$u \cdot v = \|u\| \|v\| \cos \theta \quad (\text{in } \mathbb{R}^n)$$

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